

The Impact of Utilization on Mobile Learning Applications on Academic Performance and Learning Experience Among Bachelor of Science in Information Technology Freshmen



A Project Presented to the Faculty of the
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Science, Technology, and Society
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INTRODUCTION

Background of the Study

This study focuses on how freshmen utilized mobile learning applications (m-learning apps) among Bachelor of Science in Information Technology (BSIT). As technology is thriving to satisfy human needs, freshmen are like to be early embracer of m-learning apps for educational purposes. Understanding the utilization and effects of m-learning apps on their academic performance and learning experiences is significant in enhancing educational strategies and mediums.

Past studies on m-learning apps in education have emphasize their significant impact on the learning behavior of a student (Klimova, 2019). Mobile devices have changed from means of communication to important educational resources, allowing students for opportunity and accessibility in learning. This is particularly beneficial for college students (Klimova, 2019).

Furthermore, Salhab and Daher (2023) recent literature on Nablus, Palestine shows that m-learning apps can improve student engagement and spread exchange information. Hence, there remains a gap in research specifically analyzing its effectiveness among BSIT students, especially freshmen who are adapting from high school to college. This study seeks to explore how these students utilize the use of m-learning apps and the challenges may face, and their perceptions of the overall utilization of these medium in aiding their learning process (Salhab & Daher, 2023).

However, challenges remain, especially for freshmen. Many university students in the Philippines during the COVID-19 crisis are having difficulties due to poor learning environment, unstable internet connectivity, conflict with home

responsibilities, and limited teacher scaffolds (Rotas & Cahapay, 2020), and until this present time (Naparan & Bacasmot, 2024). To effectively use of mobile learning into educational needs, institutions must address these challenges by providing adequate support and resources (Insorio, 2021). This ensures that all students, regardless of their capacity, can benefit from mobile learning technologies (Insorio, 2021).

In addition, while mobile learning application holds great potential to enhance educational outcomes, it is important to get about the associated challenges to maximize its effectiveness in college education (Magsayo, 2022).

Purpose of the Study

The purpose of this study was to identify the mobile learning application (m-learning apps) used by BSIT freshmen, as well as to see how freshmen react to its utilization like how often, what type of learning apps they used, how m-learning apps affect their academic performances, and how was their learning experience. The main variable of the study was to know how freshmen utilized the effectivity of these learning apps on their BSIT course. The researchers will collect the data by conducting a survey. The population that the researchers will conduct the study is to all BSIT freshmen students only at The University of Immaculate Conception. In addition, all data were gathered from the participants will be used for better understanding on how m-learning apps may contribute to their academic performances.

Statement of the Problem

This research aims to determine the use of mobile learning apps by each section of Bachelor of Science in Information and Technology (BSIT) freshmen students at the University of Immaculate Conception during the school year of 2024-2025. Specifically, it aims to identify the following:

1. Utilization of Mobile Learning Applications
 - 1.1 Mobile Learning Apps Used by BSIT Freshmen
 - 1.2 Frequentness of M-Learning Apps Usage
 - 1.3 Number of Hours Spent in Using M-Learning Apps
2. Academic Performance
3. Learning experiences
4. Relationship

Hypothesis

H_0 : $p\text{-value} \geq 0.05$ there is no significant difference in utilizing m-learning app when grouped according to:

- a. Academic Performance
- b. Learning Experience

H_1 : $p\text{-value} \leq 0.05$ there is significant difference in utilizing m-learning app when grouped according to:

- a. Academic Performance
- b. Learning Experience

Scope and Limitation

The coverage of this study is to deal on the utilization of mobile learning apps (m-learning apps) to the BSIT freshmen, from ages 17-23 (above). By this, students will able to reflect on their usage and to grasp the basis on the topic. Moreover, to further acquaint the students of the significance of utilizing m-learning apps.

The respondents of the study will be among BSIT freshmen students from each section, BSIT-1A and BSIT-1B, respectively, at UIC in a total of 41 respondents and their willingness and honesty to participate through face-to-face surveys. With proper netiquette while still maintaining the researcher/respondent relationship. The researchers also consider the ethical consideration since all respondents have moral and legal rights. For this study, the researchers ensured the protection of individuals by asking their permission and willingness to be one of the respondents. The researchers ensured it did not violate privacy, did not harm feelings, and all the information received was recognized and accurately represented.

Definition of Terms

To facilitate understanding of this study, various terms are defined here.

Utilization—Something is being used or can mean how much of your time is spent on a certain task.

Mobile—A portable device for connecting to the internet as accessed via smartphones or other mobile devices.

Learning apps—A digital platform that allows users to learn and acquire new skills and knowledge through electronic means.

Academic Performance—The measurement of student achievement across various subjects and defined as grades, standardized scores, and educational attainments.

Learning Experience—An interaction of any courses, program or other experience that helps a person/student learn.

Bachelor of Science in Information Technology—A degree program that typically takes four years to complete and focuses on the use of hardware and software to design, develop, and maintain IT systems.

Freshmen—A student in the first year of high school or college.

Theoretical Framework

This study was associated with combined theory of the Technology Acceptance Model (TAM) and Constructivist Learning Theory. These theories can be beneficial through examining the use of Mobile Learning Applications among BSIT freshmen and its impact on their Academic Performance and Learning Experiences. This Study also explores how these students use learning apps such as what types of M-Learning Apps, Frequency of M-Learning Apps Usage, Number of Hours Spent in Using M-Learning Apps. Guided by TAM. Furthermore, this study also suggests that students' perceptions of usefulness and ease of use influence their app adoption. Meanwhile, Constructivist Learning Theory supports the understanding of how interactive learning apps enhance engagement, how BSIT freshmen gain more knowledge, and self-directed learning.

The study examines the effects of utilizing M-Learning Apps whether there is a significant difference or no significant difference between their academic performance and learning experiences, with moderate variables such as gender, age, and section that moderate the effects by combining TAM and Constructivist Learning theory. This study aims to address the benefits and challenges of utilizing mobile learning applications in University of Immaculate Conception, particularly for BSIT freshmen.

Adapted from Granić and Marangunić (2019) & Chuang (2021)

Conceptual Framework

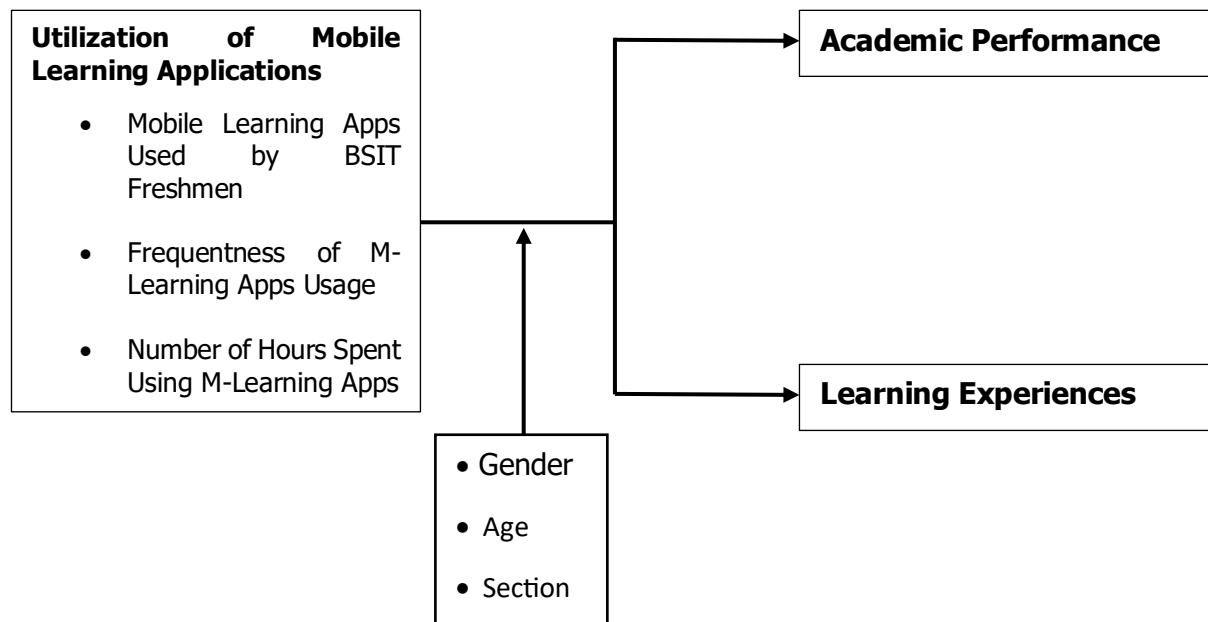


Figure 1. **Conceptual Framework of the study**

The figure above illustrates the conceptual framework used in this study. This diagram shows the relationship between Utilization of Mobile Learning, Academic Performance, and Learning Experiences among BSIT freshmen. The graph represents the procedure to identify the following. The left side represent the independent variable with indicators such as Mobile Learning Apps Used by BSIT Freshmen, Frequentness of M-Learning Apps Usage, Number of Hours Spent Using M-Learning Apps. While the right side represent the two dependent variable such as Academic Performance and Learning Experiences. Gender, Age, Section will also be moderating variables in this study. Adapted from Bermudez et al. (2022).

REVIEW OF RELATED LITERATURE

This chapter presents reviews of published books, journals, articles, and other varied sources that are viewed in relation to this study after the thorough and in-depth scrutiny done by the researchers. That may contribute to this study.

Mobile Learning Application

Mobile learning applications is a new in an education system that revise the aspect in which trend of learning takes place and the service is carry through a smartphone. This research study on mobile learning applications and its utilization among students in education and academic staff at the college (hans, G., Sidana, H., 2018).

Moreover, mobile learning is an emerging trend in many educational institutions. Mobile technology has employed wireless ways to communicate, think, learn, and share and spread of information exchange. The utilization of mobile technologies in learning and teaching can have positive conditions in education. In addition, the findings show notable social engagement themes included social-mobile interaction, building community, developing relationships, and competition. The cognitive engagement themes included attention, cognitive and meta-cognitive strategies also the results show the accessibility and ease of use of mobile learning had motivated the students (Salhab & Daher, 2023).

Investigation of utilization on m-learning apps

This study investigates the perceptions of students with respect to mobile learning services within Jordanian universities, particularly in middle of the COVID-19 pandemic. It is based on the Technological Acceptance Model (TAM) and studies apparent security, trust, risk, system quality, content quality, and service quality. The data was analyzed with Structural Equation Modelling (SEM). These results emphasized that perceived security (TAM), trust, service quality positively influence mobile learning adaption, while the impact of perceived risk showed positive effect to trust and negative affect to students' attitudes. These insights can help universities improve strategies to enhance mobile learning use (Shishakly et al., 2023).

Furthermore, Mamolo (2022) conducted a study aimed to ascertain the students' evaluation and learning experience on the developed interactive mobile application in Mathematics, the DIMaC App. The developed DIMaC application was acceptable for the students in all eight instructional material development criteria for non-print materials. Digital Interactive Math Comics (DIMaC) solicited positive experiences because the app is unique and caters to 21st century learners who are considered digital Natives. It may indicate that the way DIMaC was presented from the exciting storyline, appropriate contents, smooth interactivity, pleasant use of graphics, up-to-date information, and engaging assessment techniques were well-thought and indeed an exciting novel instructional material. Thus, the DIMaC application confirmed to the standards of instructional material development.

M-learning apps in the Philippines

De La Salle University – Dasmariñas (DLSU-D) integrates various educational apps like Flipgrid, Kahoot!, and StudyGe to enhance teaching in tourism and hospitality management courses. Faculty members perceive these educational apps as highly useful believing that they increase productivity and positively contribute to teaching performance. The apps are generally considered easy to use as faculty members find the interaction with these apps clear and understandable (Sabado, A. J., de Vera, J., Lopez, M. R., & Esplanada., 2021)

Cabaluna and Dequito (2022) was conducting a study that aims to identify the mediating effects of mobile learning system-usage- of Davao City High School students between knowledge sharing and quality of information. The results of this study were that senior high students' level of knowledge sharing was high, the level of mobile learning of mobile learning system usage suggest that the outcome was high. As a result, mobile learning systems influence students' knowledge sharing and information quality and vice versa.

Nevertheless, (Bermudez, D. M., Jaictin, F. J. O., Lauglaug, K. J. P.tamilagrosa, K. J., Piedad, B. M. A., & Jadraque, G. R., 2022) Observe the relationship between the relationship of Learner-Generated Digital Media and also the behavioral intention to use Mobile Learning Among College Students in Panabo City. The objective of the study is to examine the significant difference in the level of behavioral objective to use Mobile learning when they grouped based to gender, Age Group, and program. They

found out that there is a significant relationship between the level of Learner-Generated Digital media and Behavioral Intention to use Mobile Learning by grouped according to Gender, Age Group, Program has an effect on college students.

M-Learning apps difficulties

The insight of Romanian teachers on the advantages and disadvantages by e-learning and mobile learning application during covid-19 pandemic. Many teachers were put in the unwanted situation of being padlock due to pandemic. Therefore, e-learning was chosen as substitute for traditional learning. E-learning has proven challenge for both students and teachers (Radu, A., Ofițeru, L. G., Stoica, I., & Nedelcu, A. 2021).

In addition, Philippines also migrated to remote learning modality as a measure against covid-19 pandemic. The study reveals several difficulties as voiced out by students where analysis shows that difficulties in remote learning as mobile learning are unstable internet connectivity, insufficient learning resources, electric power interruptions; unclear learning contents, and poor learning environment. (Rotas & Cahapay, 2020).

Apart from this, Venturing the advantage and disadvantage of mobile technologies (MT) is necessary for students to ponder more knowledge and increase learning strategies. From 532 students at a state university were students mentioned

advantages and disadvantages of using mobile technology in learning activity. This study shows based on the data analysis, the main categories was referring to the benefits and difficulties on using MT. the main categories with regards to benefits are: information using mobile technologies, efficient communication, opportunities for educational process. With regards to difficulties there was technological difficulties, damage information, diminishing communication a socialization ability, negative effects on the educational process, ecological and ethical disadvantages (Măță & Dobrescu, 2022).

METHODOLOGY

This chapter presents the explanation about procedures used in this study to find out the results to the gathered data from respondents. This section includes research locale, research respondents, research design, and research instrument.

Research Locale

The study coverage was conducted at University of Immaculate Conception (UIC) located at Father Selga St., Davao City. The school was a private university that provided most of college courses here in Davao. The study will focus on BSIT freshmen students from each section, BSIT-1A and BSIT-1B, respectively, with a total of 28 students/respondents who participated willfully. Furthermore, the accuracy of the gathered data may depend on their honesty for the research and contribute to a better understanding.



Figure 2. **Map of the Place of study**

Research Respondents

Using the purposive sampling, participants are selected or sought based on pre-selected criteria using survey questionnaire and adaptive 5-point Likert Scale from Bermudez et al. (2022). The respondents will ask by how frequent and how many hours per week they spend measured by much less, less focus, maintain, more focus, much more focus; less than 1 hour, 1-2 hours, 2-3 hours, 4 hours, more than 4 hours and what m-learning apps are used by BSIT freshmen measured by e-books, educational games, Language Learning apps, Puzzle games for education, others and interactive quizzes, Problem solving, Video tutorials, Discussion, Others and their learning experience will be assessed based on student satisfaction, engagement, and perceived effectiveness. BSIT freshmen was chosen as our respondents and primary data for statistical analysis for better understanding the effect of utilization of m-learning apps.

Research Design

This study used correlational research. This utilized the relationship between it uses statistical analysis. However, it mostly observational in terms of data collection. A correlational exist when one variable increase or decreases correspondingly with another variable. Here, variables are not manipulated (Bloomfield & Fisher, 2019). Nevertheless, quantitative research was use to observe the relationship between variables. Pearson r used to determine the relationship between three quantitative variables on which the three variables relate to one another. This tool will be used to measure the significant correlation between the utilization of mobile learning applications, academic performance, and learning experiences.

Research Instrument

The coverage of this study is to focus on utilization of m-learning apps among BSIT freshmen at University of the Immaculate Conception and to gather data from each section and ensure their privacy remains confidential with the use of survey questionnaire. The duration of survey will last within a couple of minutes. In this study will be measured using IBM SPSS and adapted 5-point Likert scale by Rensis Likerts' from Bermudez et al. (2022). by getting the mean and explain the null hypothesis by running statistical test.

The strategic thinking skills of the students/respondents will be categorized as describe following Rensis Likerts' scale:

Table 1. Scale for Utilization of Mobile Learning Applications, Academic Performance, Learning Experiences

Scale	Range of means	Verbal Description	interpretation
5	4.50-5.00	Very Satisfied	It indicates that the utilization of m-learning application has met the expectations.
4	3.50-4.49	Satisfied	It indicates that the utilization of m-learning applications has met the expectations.
3	2.50-3.49	Neither satisfied/dissatisfied	It indicates that the m-learning applications has met neither on the borderline of what was expected.
2	1.50-2.49	Dissatisfied	It indicates that the m-learning applications wasn't met the expectations.
1	1-1.49	Very Dissatisfied	It indicates that the m-learning applications was unsatisfactory.

This table shows the scales for Utilization of Mobile Learning Applications, Academic Performance, Learning Experiences. It also consists of the range of mean, descriptive rating, and descriptive interpretation. Adapted from Bermudez et al. (2022).

Results and Discussion

Results

This chapter represents the findings of the data that the researchers gathered from the survey, below are the mathematical and arithmetic computations and this was also adapted from specified source.

Table 2. **Demographic profile of the respondents**

		Frequency	Percentage
Gender	Male	23	82.1
	Female	5	17.9
Age Grouped	17-18 years old	19	67.9
	19-20 years old	9	32.1
Section	BSIT-1A	11	39.3
	BSIT-1B	17	60.7

As shown in Table, there is a total of 28 respondents in this study with 82.1% are male and 17.9% are female. There were 67.9% of 17-18 years old and 32.1% of 19-20 years old. 39.3% of the respondents came from BSIT-1A while there are 60.7% of respondents came from BSIT-1B. This also used by Bermudez et al. (2022).

SOP1: **Mobile Learning Apps Used by BSIT Freshmen**

Table 3. **Mobile Learning Applications Used**

		Frequency	Valid Percent
Valid	E-books	3	10.7
	educational games	8	28.6
	language learning app	8	28.6
	others	9	32.1
	Total	28	100.0

The table shows the M-Learning Application utilized by BSIT freshmen. The most popular applications are educational games and language learning apps having a both of 28.6% of respondents. E-books, on the other hand, are the least used with only of 10.7% respondents utilizing them. This suggests a desire for interactive and engaging learning experiences over traditional learning materials or text-based formats. The remaining 32.1% of respondents use other unspecified M-Learning Applications, suggesting the other variety available tools for learning preferences. Adapted from Bermudez et al. (2022).

SOP1: Mobile Learning Apps Used by BSIT Freshmen

Table 4. **Specific Features of Mobile Learning Applications that Respondents Find Useful**

		Frequency	Valid Percent
Valid	interactive quizzes	6	21.4
	problem solving	1	3.6
	video tutorials	15	53.6
	discussion	4	14.3
	others	2	7.1
	Total	28	100.0

Based on the table the specific features of M-Learning Applications that respondents find useful are video tutorials which were used by (15)53.6% of respondents. Interactive quizzes also consider find useful by (6)21.4% of respondents. Nevertheless, other features that were deem useful include discussion (14.3%), problem-solving (3.6%), and others (7.1%). These findings suggest that respondents chose specific features that enhance their learning experiences and provide opportunities to gain more knowledge. Adapted from Bermudez et al. (2022).

SOP1: **Frequentness of M-Learning Apps Usage**

Table 5. **Frequency of Mobile Learning Application Used for Studies**

	Much less	Less focus	Maintain	More Focus	Much More Focus	Mean
How often do you use mobile learning applications for studies			19	9		2.6786
Total	28					

The table shows how frequent respondents utilized M-Learning Applications for studies. There are 19 respondents' states that they maintain the used of M-Learning apps for studies. Meanwhile, there are also 9 respondents were more focus on utilizing M-Learning apps for their studies. Therefore, the overall results are conveyed into mean score with average of 2.67. It indicates that the M-Learning Applications has met neither on the borderline of what was expected. Adapted from Bermudez et al. (2022).

SPO1: Number of Hours Spent in Using M-Learning Apps

Table 6. Average Time spent per week on M-Learning Applications

	N	Mean
Less than 1 hour	2	3.3929
1-2 hours	9	
2-3 hours		
4 hours	10	
More than 4 hours	7	
Total	28	

The table represent insights into the average time spent per week on M-Learning Applications by the respondents. A small number (2) of respondents said that they spending less than 1 hour per week on utilizing M-Learning Apps. 9 respondents, on the other hand, state that they spent 1-2 hours per week on utilizing M-Learning Apps. 10 respondents spent 4 hours by utilizing M-Learning Apps per week. Nevertheless 7 respondents state that they spent more than 4 hours on utilizing M-Learning Apps per week with average mean score of 3.3929. It indicates that the m-learning applications has met neither on the borderline of what was expected. Adapted from Bermudez et al. (2022).

SOP2: Academic Performance

Table 7. **Influence of M-Learning Application on the academic performance of BSIT freshmen**

	Very Poor	Poor	Fair	Good	Very Good	Mean
How often do you complete assignments on time?			8	14	6	3.9286
How do you rate your performance in your recent exams?		1	17	8	2	3.3929

The table indicates that the use of M-Learning Applications has influence their academic performance based on their ability on how often they complete assignments on time and their performance in recent exams. According to the data there were 8 of respondents rated their ability to complete assignments as "Fair". Meanwhile, 14 of respondents rated themselves as "Good" on completing their assignments on time. Furthermore, 6 respondents rated their ability as "Very Good" to complete assignments on time with having an overall mean score of 3.9286. It indicates that the utilization of m-learning applications has met the expectations.

Furthermore, for their recent exam performance, 1 respondent rated as "Poor", while the majority has 17 of respondents rated as "Fair" and also 8 of respondents rated their recent exam performance as "Good". Only 2 respondents considered their performance as "Very Good". Lastly the mean score of their ratings on recent exam performance has 3.3929. It indicates that the m-learning applications has met neither on the borderline of what was expected. Adapted from Bermudez et al. (2022).

SOP3: Learning Experience

Table 8. **Learning Experience with M-Learning of BSIT Freshmen**

	Very Dissatisfied	Dissatisfied	Neither satisfied/ dissatisfied	Satisfied	Very Satisfied	Mean
How satisfied are you with the mobile learning applications you use?	1			22	5	4.0714
How effective do you find when using mobile learning applications?	1	1		21	5	4.0000
How effective do you find mobile learning applications in helping you understand the course material?	1	2	1	20	4	3.8571

The table shows the learning experience of BSIT freshmen using M-Learning Application. When researcher asked about their satisfaction with M-Learning Applications, 22 respondents rated themselves "Satisfied", while 5 respondents rated themselves "Very Satisfied". Only 1 respondent expressed "Dissatisfaction". The Overall mean score for this category has 4.0714. It indicates that the utilization of m-learning applications has met the expectations.

As for the effectiveness of M-Learning Apps, 21 respondents found that "Satisfied" or found effective and 5 respondents found M-Learning Apps very effective or "Very Satisfied". Meanwhile, 1 respondent rate it as "Dissatisfied", and another 1 respondent was "Very Dissatisfied" with the apps' effectiveness. The mean score for this category has 4.0000. It indicates that the utilization of m-learning applications has met the expectations.

Lastly, in terms of how effective M-Learning Apps help students understand course material, 20 respondents rated their experience as "Satisfied" and found it effective, while 4 respondents rated their experience as "Very Satisfied" and found it very effective. However, 1 student remain neutral or "Neither satisfied/ dissatisfied" and 3 respondents rated their experiences as "Very Dissatisfied" or "Dissatisfied". The mean score for this category has 3.8571. It indicates that the utilization of m-learning applications has met the expectations. Adapted from Bermudez et al. (2022).

Table 9. **ANOVA Distribution of respondents' Significant Difference in the Academic Performance to Gender, Age, Section, and Program ANOVA (Between Group)**

		ANOVA				
		Sum of Squares	df	Mean Square	F	Sig.
Gender	Between Groups	.625	5	.125	.790	.568
	Within Groups	3.482	22	.158		
	Total	4.107	27			
Age	Between Groups	1.837	5	.367	1.894	.136
	Within Groups	4.270	22	.194		
	Total	6.107	27			
BSIT -1A, BSIT -1B	Between Groups	1.500	5	.300	1.274	.310
	Within Groups	5.179	22	.235		
	Total	6.679	27			

Interpretation: There is no sufficient evidence to say that there is a statistically significant difference between groups as determined by on-way ANOVA. For gender there is no significant difference ($F(5,22)=.790, p=0.568$) on how m-learning apps affects their academic performance. For age there is no significant difference ($F(5,22)=1.894, p=0.136$) on how m-learning apps affects their academic performance. For section there is no significant difference ($F(5,22)= 1.274, p=0.310$) on how m-learning apps affects their academic performance. For section there is no significant difference. Adapted from Bermudez et al. (2022).

SOP4: **Relationship**Table 10. **Relationship Between the Utilization of mobile learning applications, Academic performance, and Learning experiences**

		Utilization of mobile learning applications	Academic performance	Learning experiences
Utilization of mobile learning applications	Pearson Correlation	1	.346	-.196
	Sig. (2-tailed)		.071	.318
	N	28	28	28
Academic performance	Pearson Correlation	.346	1	-.263
	Sig. (2-tailed)	.071		.177
	N	28	28	28
Learning experiences	Pearson Correlation	-.196	-.263	1
	Sig. (2-tailed)	.318	.177	
	N	28	28	28

Based on the table, the Pearson product-moment correlation was used to determine the relationship between the Utilization of mobile learning applications, Academic performance, and Learning experiences. The results show There is a moderate positive correlation ($r = 0.346$) between Utilization of mobile learning applications and academic performance which mean the p-value (0.071) is greater than 0.05, suggesting that this correlation is not statistically significant at the conventional level. However, it is close to significance, indicating a potential difference worth exploring further. There is a weak negative correlation ($r = -0.196$) Utilization of mobile learning applications and learning engagement it means that the p-value (0.318) indicates that this correlation is not statistically significant. There is a weak negative correlation ($r = -0.263$) between academic performance and learning experiences which means that p-value (0.177) suggests this correlation is also not statistically significant. Since the p-value is $0.05 > 0.05$, then we reject the alterative hypothesis. Therefore, there is no significant relationship between the Utilization of mobile learning applications, Academic performance, and Learning experiences. Adapted from Bermudez et al. (2022).

Conclusion and Recommendation

Conclusion

The study looks to investigate on utilization of Mobile Learning Application among BSIT freshmen in University of Immaculate Conception. The results revealed that utilizing modern mediums for education has no sufficient evidence to say that there is a statistically significant difference between groups as determined by on-way ANOVA for gender, age, their section and the relationship between the utilization of mobile learning application, academic performance, and learning experience it means that it is fine to use M-Learning Apps and it also fine to not utilize M-Learning Apps. Since the p-value is $0.0005 > 0.05$, therefore, there is no significant relationship between the utilization of mobile learning applications, academic performance, learning experiences. However, the results show moderate positive correlation that indicating a potential difference that may worth to explore further.

Recommendation

- **Increase Support for Mobile Learning:** Since students find M-Learning Apps useful but do not heavily rely on them, the university could promote the wider adaption of these tools by integrating them more deeply into the curriculum. Providing structured guidance on how to maximize their potential could improve engagement and outcomes.
- **Address Challenges:** Based on past literature, issues such as unstable internet connectivity and a lack of support might difficult to use of those tools. Ensuring that students have the necessary configuration and assistance will help increase their usage and benefit.
- **App Development:** Given that certain apps like e-books and quizzes were more commonly used, the university could focus on developing or adapting more tailored apps that cater specifically to the needs of BSIT students, such as coding simulators or interactive tech tutorials.

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Appendices

Appendix A

The University of the immaculate conception

A Case Study on The Utilization of Mobile Learning Applications Among BSIT Freshmen

Praised be Jesus and Mary!

We humbly invited you to participate in this survey, which aims to collect insights on how BSIT freshmen in the University of the Immaculate Conception utilize their mobile learning apps. **By completing this questionnaire, you formally acknowledge and accept the terms and conditions of participation.** Be assured that all information you provide will remain confidential and will not be publicly disclosed. Your participation is highly valued and appreciated. Thank you!

Demographic Information:

Date: _____

Section: _____

Demographic profile

Age	
17-18	
19-20	
21-22	
23 and above	
Gender	
Male	
Female	

Utilization of Mobile Learning Applications:

1.1 Mobile Learning Apps Used by BSIT Freshmen

	e-books	Educational games	Language learning app	Puzzle games for education	Others
Which types of mobile learning applications do you use?					
	interactive quizzes	Problem solving	Video tutorials	Discussion	others
What specific features of mobile learning applications do you find most useful?					

1.2 Frequentness of M-Learning Apps Usage

	Much less	less focus	maintain	more focus	much more focus
How often do you use mobile learning applications for your studies?					

1.3 Number of Hours Spent in Using M-Learning

	Less than 1 hour	1-2 hours	2-3 hours	4 hours	More than 4 hours
On average, how many hours per week do you spend using mobile learning applications?					

Academic Performance:

	Very poor	Poor	Fair	Good	Very good
How do you rate your performance in your recent exams?					
How often do you complete assignments on time?					

Learning Experiences:

	Very dissatisfied	Dissatisfied	Neither satisfied/dissatisfied	Satisfied	Very Satisfied
How satisfied are you with the mobile learning applications you use?					
How effective do you find when using mobile learning applications?					
How effective do you find mobile learning applications in helping you understand course material?					

Appendix B

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Educational Background:

Undergraduate	University of Immaculate Conception Father Selga St., Davao City	Present
Secondary	Communal National High School Purok 3, Barangay Communal, Davao city	2017-2024
Elementary	Buhangin Central Elementary School Kilometer 6, Manuel M. Garcia Road, Barangay Buhangin, Davao City	2011-2016

Work experience:

STI College Davao
J.P. Laurel Avenue corner Acacia, Davao City, 8000 Davao del Sur
Work Immersion Student
Part time computer technician

Awards/Affiliations

Conduct awardee
With honors

Trainings and Seminars Attended

Calamity and Disaster Preparedness
Drug Education
Environmental Education
Symposium on Animal Welfare to Care for the Creation of the Universe
Drug Situationer

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Educational Background:

Undergraduate	University of Immaculate Conception Father Selga St., Davao City	Present
	International School of Technology, Arts and Culinary Cor. Loyola Street J.P Laurel Ave. Davao City	2022-2024
Secondary	Antipas National High School Poblacion Antipas, North Cotabato	2016-2020
Elementary	Kamasi Elementary School Brgy. Camasi Pres. Roxas, North Cotabato	2011-2016

Work experience:

Kairos Kitchen (On the Job Training)	December 2022 – January 2023
Panadero (On the Job Training)	June 2023 – August 2023
D' Leonor Hotel (On the Job Training)	August 2023 – September 2023

Awards/Affiliations

ISTAC Culinary Olympics 2023 (Grand Winner) – Celebration Cake

Trainings and Seminars Attended

Calamity and Disaster Preparedness
Drug Education
Environmental Education
Symposium on Animal Welfare to Care for the Creation of the Universe
Drug Situationer